



QRT Inc.

Reliability Test Report

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COMPANY : UBIVELOX Inc.

PRODUCT : UONE-A10

PACKAGE : 8DFN

PURPOSE : Qualification

ISSUED DATE : 2026-03-04

REFERENCE No. : 2509-15548

Prepared By : Gangwhoan Lim

Approved By : Kwangsun Jung

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1 Qualification Plan and Results

The purpose of these tests is evaluation (or monitoring) of **UONE-A10**.

Test conditions are specified in customers in-house test plan and all test procedures comply with **JEDEC** standards.

Obligation to perform exact procedure to reference documents is QRT Inc. responsibility only, but establishing failure criteria and judgment of pass/fail is customer's responsibility.

Test Items	Test Condition	Duration	Sample Size	Failed Unit ¹⁾	Test Method ²⁾
Precondition	BAKE : 125 (+5, -0) °C SOAK : 30 °C ± 2 °C, 60 % ± 3 % R.H. REFLOW : ≥ 260 °C	24 hrs 192 hrs 3 Cycles	120 ea	0	JESD22-A113I:2020 (PRECONDITIONING OF NONHERMETIC SURFACE MOUNT DEVICES PRIOR TO RELIABILITY TESTING)
High Temperature Operating Life	125 °C ± 5 °C V0 = 5.5 V	1 000 hrs	80 ea	0	JESD22-A108G:2022 (TEMPERATURE, BIAS, AND OPERATING LIFE)
High Temperature Storage	150 (+10, -0) °C	1 000 hrs	45 ea	0	JESD22-A103E.01:2021 (HIGH TEMPERATURE STORAGE LIFE)
^ Unbiased-Highly Accelerated Stress Test	130 °C ± 2 °C, 85 % ± 5 % R.H. / 230 kPa	96 hrs	50 ea	0	JESD22-A118B.01:2021 (ACCELERATED MOISTURE RESISTANCE -UNBIASED HAST)
^ Temperature Cycling	-65 (+0, -10) °C ~ 150 (+10, -0) °C	500 Cycles	45 ea	0	JESD22-A104F.01:2023 (TEMPERATURE CYCLING)
Solderability [Dip & Look]	Preconditioning : Condition E Dry Bake : 155 °C Exposure time : 4 hours ± 15 min Flux Immersion Time : 5 - 10 sec Solder Temperature : 245 ± 5 °C Solder Immersion Time : 5 +0/-0.5 sec Solder Immersion speed : 25 ± 6 mm/sec Solder Immersion depth : 0.5 ± 0.1 mm Solder Dipping angle : Horizontal	-	15 ea	0	J-STD-002D:2011 (Solderability)
Electrostatic Discharge (ESD) Human Device Model	± 2 000 V	-	9 ea	0	JS-001-2017
Electrostatic Discharge (ESD) Charged Device Model	± 500 V	-	9 ea	0	JS-002-2018
Latch-Up	Signal Pin Test : ± 100 mA Supply Test : + 8.25 V (Test Temp : Room)	-	6 ea	0	JESD78F:2022

Additional Requirement : N/A

Note :

1) Test results are based on the "Final Test Result" provided by customer.

2) Although the name of test item is same, reference documents can be JEDEC.

^ "Preconditioning" was applied before reliability test.

3) "This laboratory is not accredited for the test results " * " marked."

2 Test Vehicles

Manufacturer : UBIVELUX Inc.

Part Number : UONE-A10

Description : -

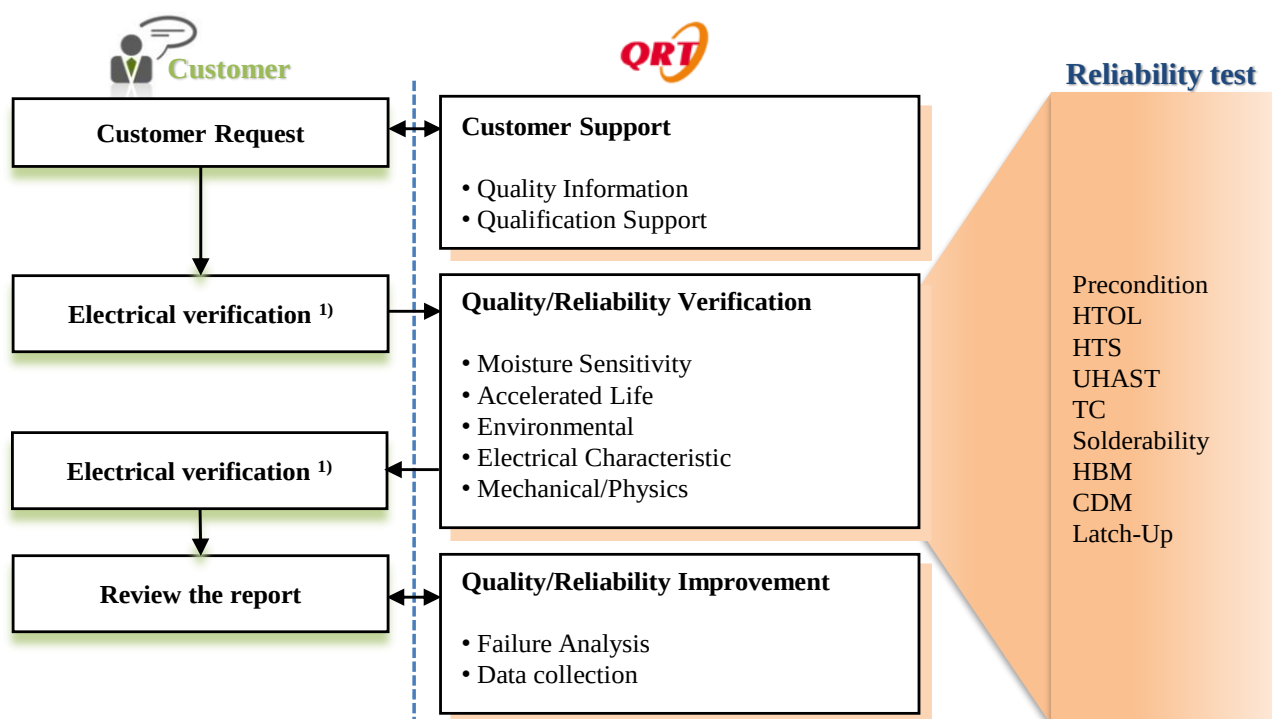
Package Type : 8DFN

Ass'y Site : -

Fab Site : -

3 Test Flow/Procedure

All reliability tests performed by QRT Inc. inc. comply with international standard. QRT inc. provide customer support featuring easy-understanding documents & technical information. And product monitoring activities to confirm quality and reliability level by sampling inspection and reliability test.



NOTE ¹⁾ Customer's Responsibilities only. Electrical Test can be omitted depending on customer's request

Figure 3-1. Reliability Test Flow

4 Preconditioning (including MSL)

Preconditioning of SMD packages is used to simulate the effects of board assembly on moisturized packages, prior to reliability testing. This allows reliability testing at the component level on as shippable products with a board assembly simulation. During preconditioning, test samples are subjected to temperature cycling (optional), dry bake, moisture soaking, solder reflow simulation, flux, rinse, dry, and electrical test before reliability testing.

4.1 Preconditioning Procedure

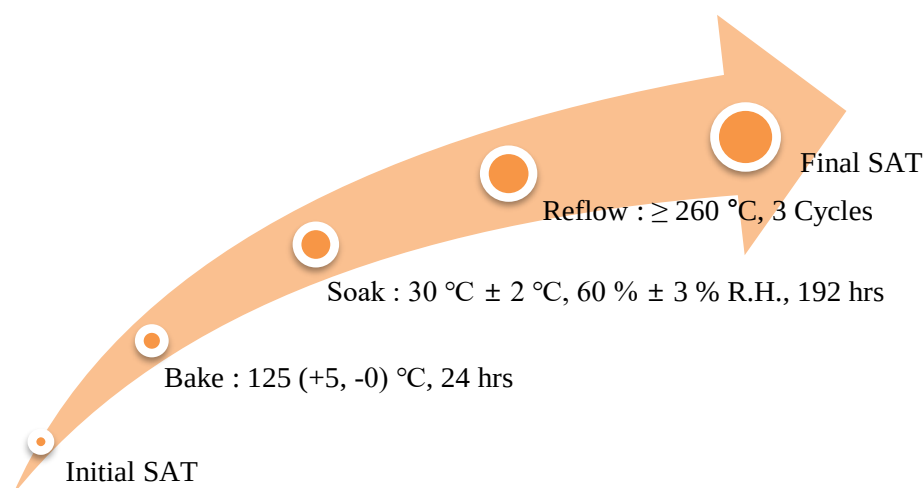


Figure 4-1. Preconditioning Flow

4.2 Reflow Profile

Reflow Profile was set using 3 representative components. All temperatures refer to the center of the package, measured on the package body surface that is facing up during assembly reflow.

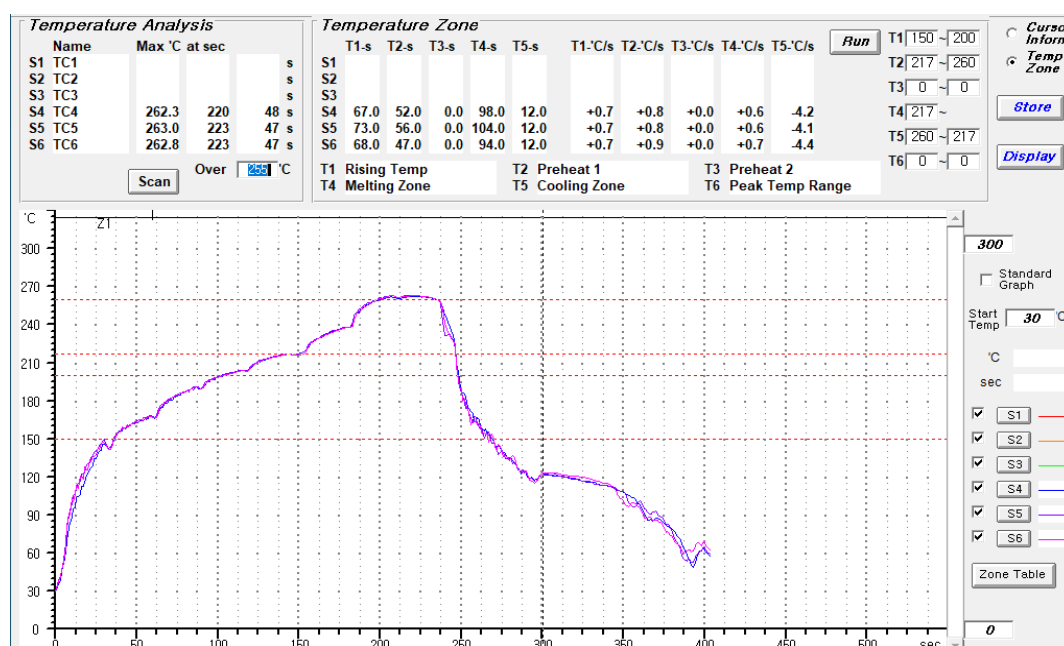


Figure 4-2. Reflow Profile



5 SAT Analysis

5.1 SAT Inspection

Sample	Defect List	Result				Comments
		Yes	No	N/A	If Yes	
					Location	
Initial	Delamination		○		Die surface to molding	
					Mold to lead frame(paddle)	
					Die to adhesive tape	
					Die to substrate	
					Molding to substrate	
					Others:	
	Crack		○		Chip	
					Molding compound	
					Substrate	
					Others:	
	Others					
After Reliability Test	Delamination		○		Die surface to molding	
					Mold to lead frame(paddle)	
					Die to adhesive tape	
					Die to substrate	
					Molding to substrate	
					Others:	
	Crack		○		Chip	
					Molding compound	
					Substrate	
					Others:	
	Others					
Comment : -						

5.1 X-ray Inspection (cont'd)

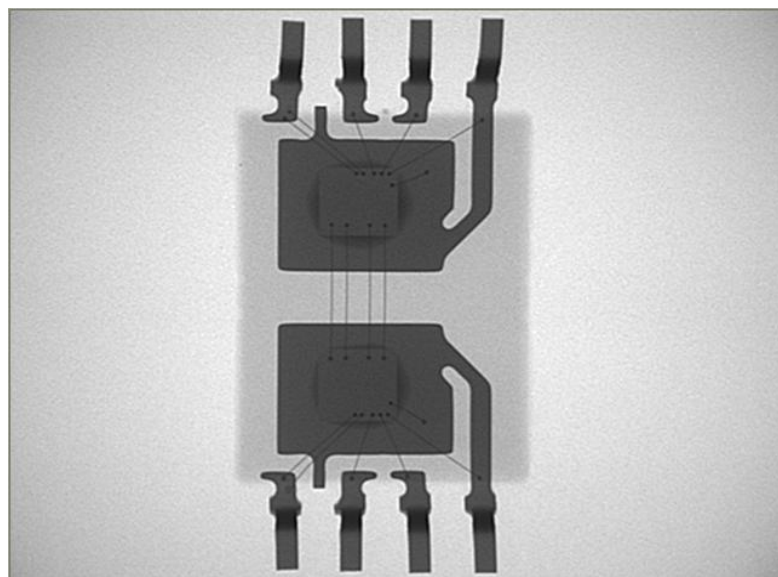


Figure 5-1. X-Ray View (Topside)



Figure 5-2. X-Ray View (Tilted)

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

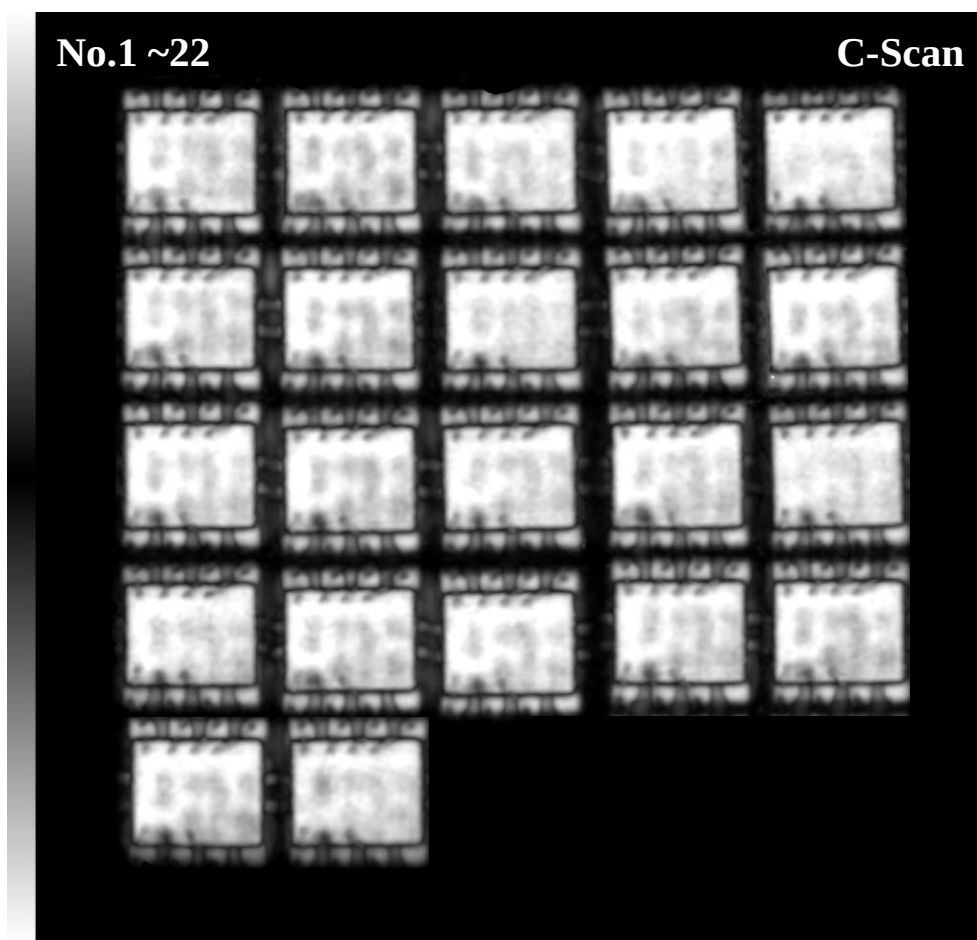


Figure 5-3. SAT images before Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

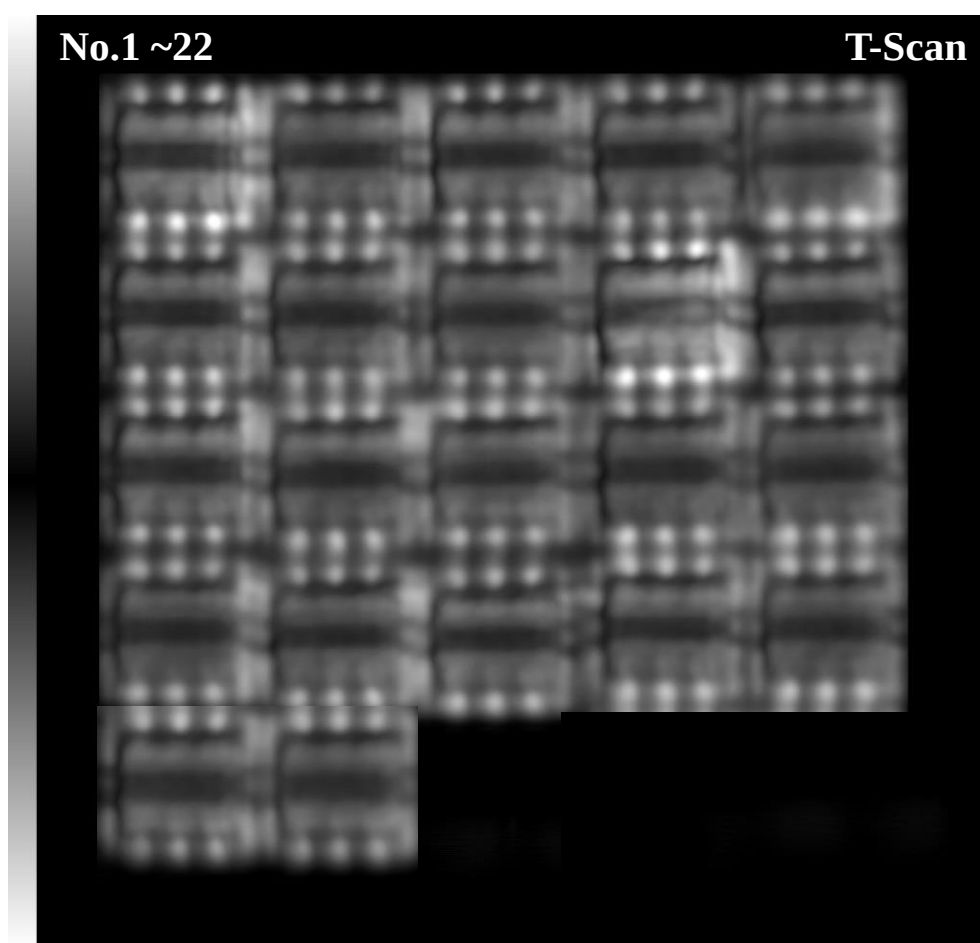


Figure 5-4. SAT images before Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object. It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

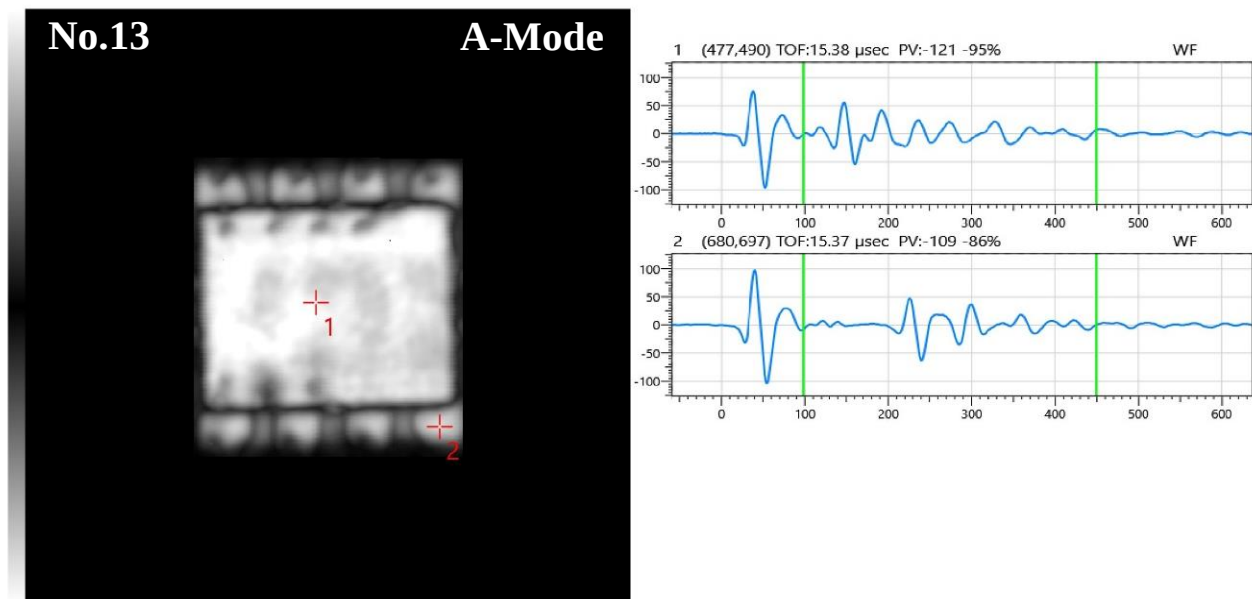


Figure 5-5. SAT Waveform Data before Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

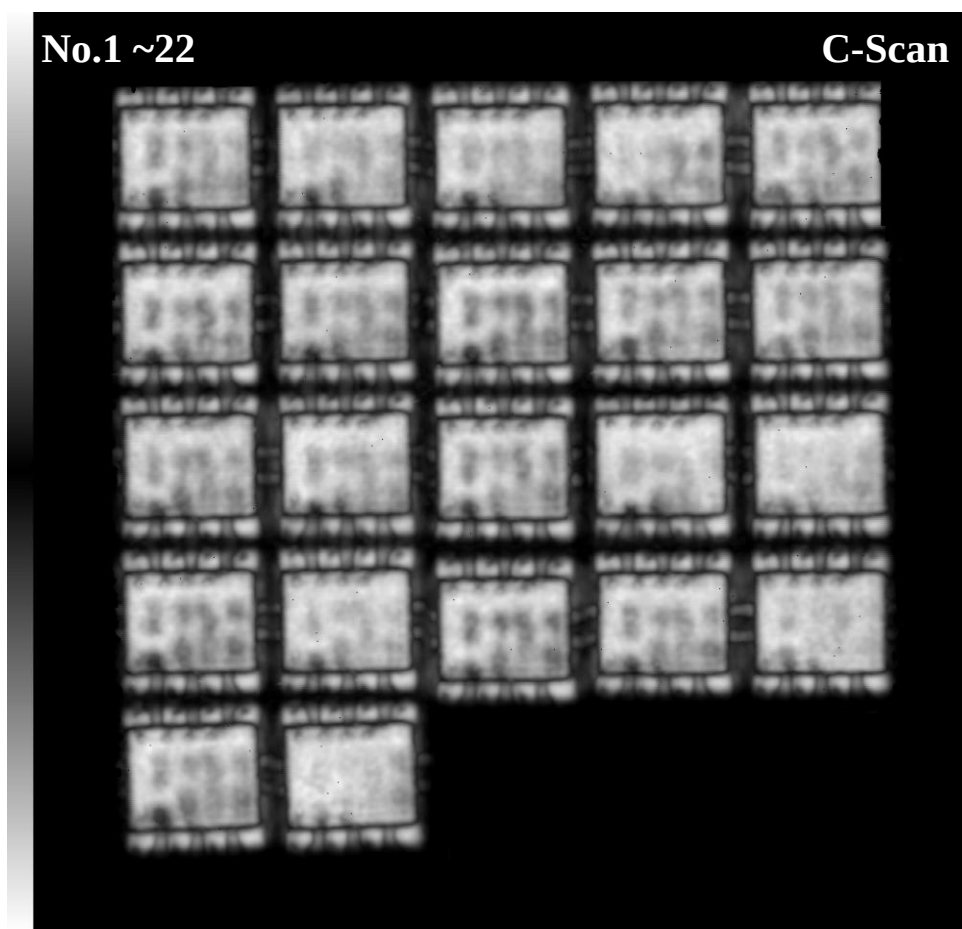


Figure 5-6. SAT images after Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

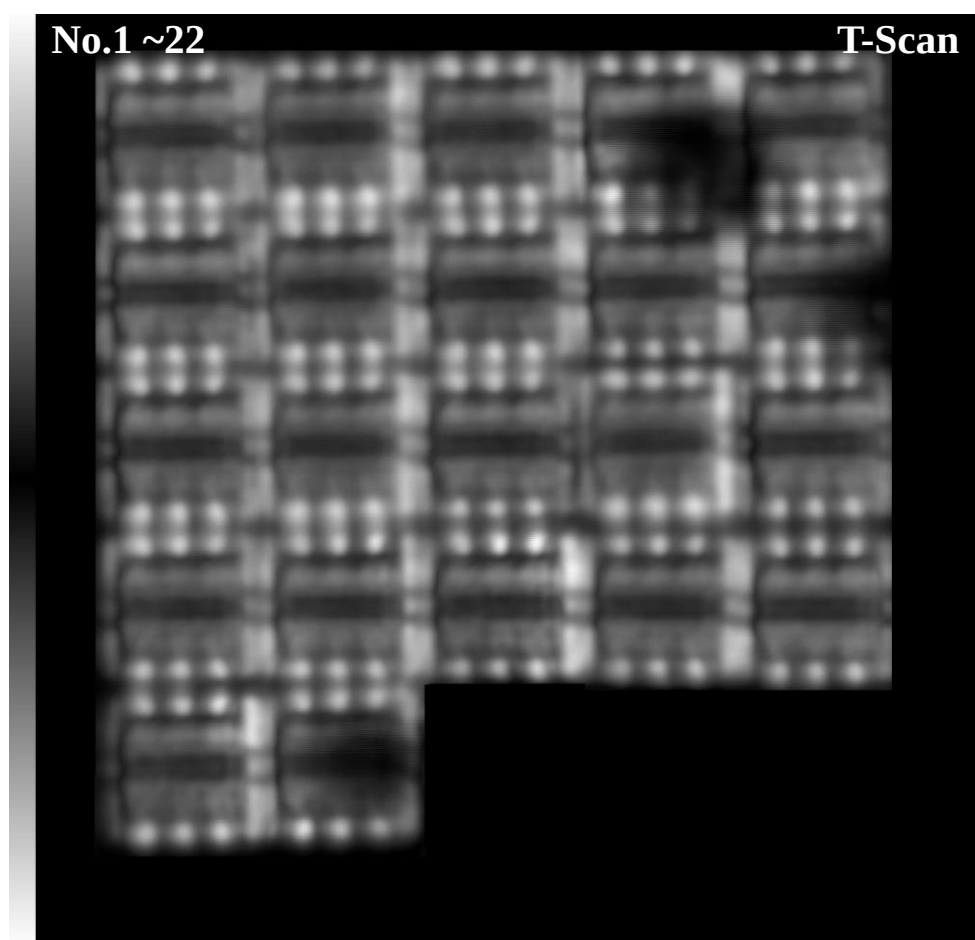


Figure 5-7. SAT images after Precondition

5.1 SAT Inspection (cont'd)

A Scanning Acoustic Tomograph (SAT) is a device which uses focused sound to investigate, measure, or image an object . It is commonly used in failure analysis and non-destructive evaluation to detect voids, cracks and delaminations within microelectronic package.

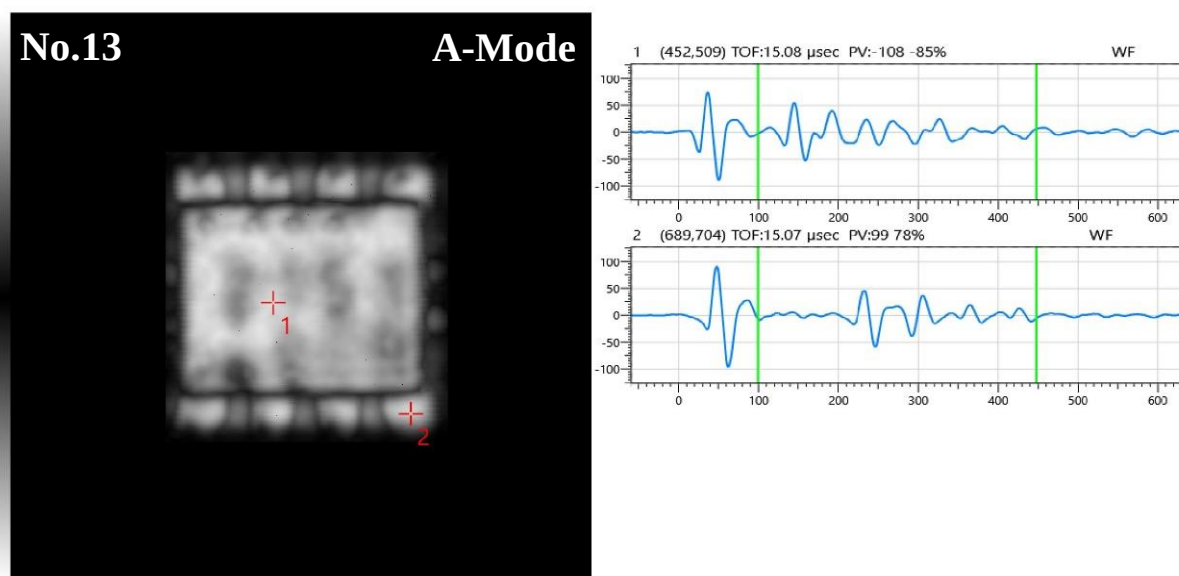


Figure 5-8. SAT Waveform Data after Precondition

6 Reliability Tests

6.1 High Temperature Operating Life (HTOL)

The High Temperature Operating Life (HTOL) or steady-state life test is performed to determine the reliability of devices under operation at high temperature conditions over an extended period of time. It consists of subjecting the parts to a specified bias or electrical stressing, for a specified amount of time, and at a specified high temperature.

Test condition

Chamber condition : $125\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$

Bias configuration : $V_0 = 5.5\text{ V}$

Duration : 1 000 hrs

Sample Size : 80 ea

6.2 High Temperature Storage (HTS)

This test is typically used to determine the effects of time and temperature, under storage conditions, for thermally activated failure mechanisms and time-to failure distributions of solid state electronic devices, including nonvolatile memory devices (data retention failure mechanisms). Thermally activated failure mechanisms are modeled using the Arrhenius Equation for acceleration. During the test, accelerated stress temperatures are used without electrical conditions applied.

Test Condition

Chamber condition : $150\text{ }(^{+10}, -0)\text{ }^{\circ}\text{C}$

Duration : 1 000 hrs

Sample Size : 45 ea

6.3 Unbiased-Highly Accelerated Stress Test (UHAST)

This test is performed for the purpose of evaluating the reliability of non-hermetic packaged solid-state devices in humid environments. It is a highly accelerated test which employs temperature and humidity under non-condensing conditions to accelerate the penetration of moisture through the external protective material (encapsulant or seal) or along the interface between the external protective material and the metallic conductors which pass through it. Bias is not applied in this test to ensure the failure mechanisms potentially overshadowed by bias can be uncovered (e.g., galvanic corrosion).

Test Condition

Chamber condition : 130 °C $\pm$ 2 °C, 85 % $\pm$ 5 % R.H. / 230 kPa

Duration : 96 hrs

Sample Size : 50 ea

6.4 Temperature Cycling (TC)

This test is conducted to determine the ability of components and solder interconnects to withstand mechanical stresses induced by alternating high- and low-temperature extremes. Permanent changes in electrical and/or physical characteristics can result from these mechanical stresses.

Test Condition

Chamber condition : -65 (-0, +10) °C ~ 150 (-10, +0) °C

Duration : 500 Cycles

Sample Size : 45 ea

7. ESD Test Results

7.1 HBM Test Results

Pin Combinations Sets		No. of Zaps (Polarity)	Test Level	Sample Size	Test Result ¹⁾	Classification
Test Pin (Terminal A)	Ref. Ground (Terminal B)					
VSS, I/O	VDD	1 (±)	2 000 V	3 ea	Pass	2
VDD, I/O	VSS		2 000 V	3 ea		
Every I/O	All I/O s (except for test pin)		2 000 V	3 ea		
NOTE						
1) Functional test conducted by customer site - Reference Document : JS-001-2017						

7.2 CDM Test Results

Charge Method (Dielectric)	Discharge Pin	No. of Zaps (Polarity)	Test Level	Sample Size	Test Result ¹⁾	Classification
Field Induced (FR-4)	Every pin (except for N/C)	1 (±)	500 V	9 ea	Pass	C2a
NOTE 1) Functional test conducted by customer site - Reference Document : JS-002-2018						

8. Latch-up Test Results

Stress Type (Test Method) (Temperature)	Test Pin or Pin Group	Condition of untested input pins	(Polarity) Stress Level	Sample Size	Test Result		Immunity Level
					Latch-up Detection ¹⁾	F/T ²⁾	
Signal Pin Test (I-Test) (Room)	I/O	Maximum pin Operating Voltage & Minimum pin Operating Voltage	(±) 100 mA	3 ea	Pass	Pass	A
Supply Test (Room)	Power		(+) 8.25 V	3 ea	Pass	Pass	
NOTE							
1) Latch-up Detection Criteria							
a. If absolute Inom ≤ 25 mA, then absolute Inom + 10mA							
b. If absolute Inom > 25 mA, then absolute Inom × 1.4							
2) Functional test conducted by customer site							
- Reference Document : JESD78F:2022							

9. Descriptions of ESD (ElectroStatic Discharge)

Electrostatic Discharge, or ESD, is a single-event, rapid transfer of electrostatic charge between two objects, usually resulting when two objects at different potentials come into direct contact with each other. ESD can also occur when a high electrostatic field develops between two objects in close proximity. ESD is one of the major causes of device failures in the semiconductor industry.

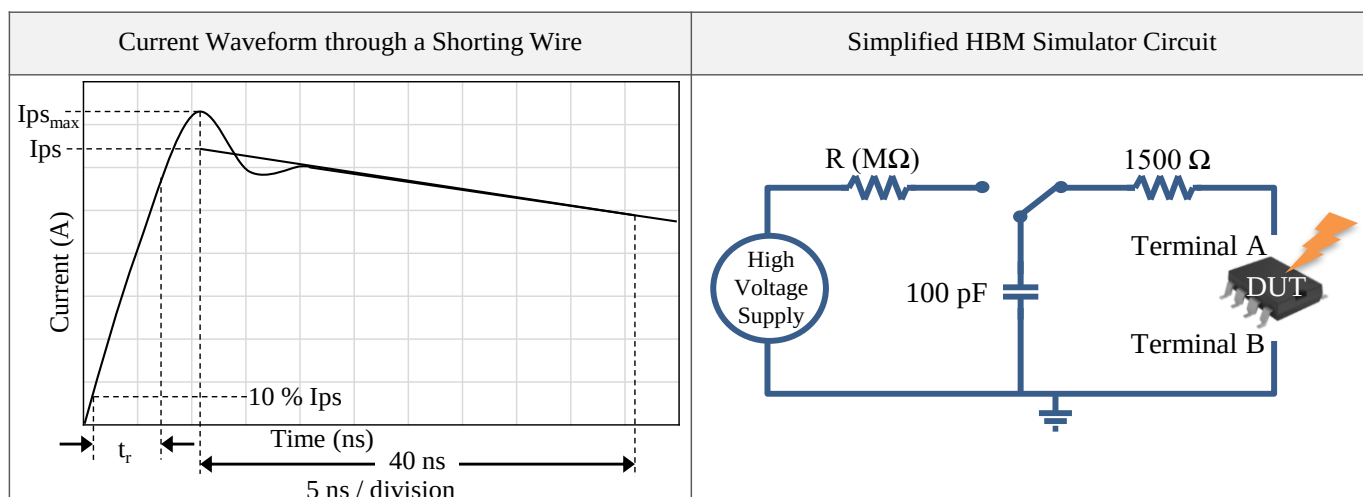
9.1 HBM (Human Body Model)

This test is classifying microcircuits according to their susceptibility to damage or degradation by exposure to a defined electrostatic Human Body Model (HBM) discharge (ESD).

9.1.1 HBM ESD Component Classification Levels

Classification	Voltage Range (V)
0A	< 125 V
0B	125 to < 250 V
1A	250 to < 500 V
1B	500 to < 1000 V
1C	1000 to < 2000 V
2	2000 to < 4000 V
3A	4000 to < 8000 V
3B	≥ 8000 V

9.1.2 HBM Current Waveform & Simplified HBM Simulator Circuit



9.1.3 HBM Tester

Tester Model	Maker	Serial Number	Effective term of Calibration	Accredited Calibration Lab.
MK2	Thermo Fisher Scientific	401208	2025.10.22 ~ 2026.10.22	SICT(KC01-018)
MK4. TE	Thermo Fisher Scientific	1806237	2025.10.22 ~ 2026.10.22	

9. Descriptions of ESD (ElectroStatic Discharge)

Electrostatic Discharge, or ESD, is a single-event, rapid transfer of electrostatic charge between two objects, usually resulting when two objects at different potentials come into direct contact with each other. ESD can also occur when a high electrostatic field develops between two objects in close proximity. ESD is one of the major causes of device failures in the semiconductor industry.

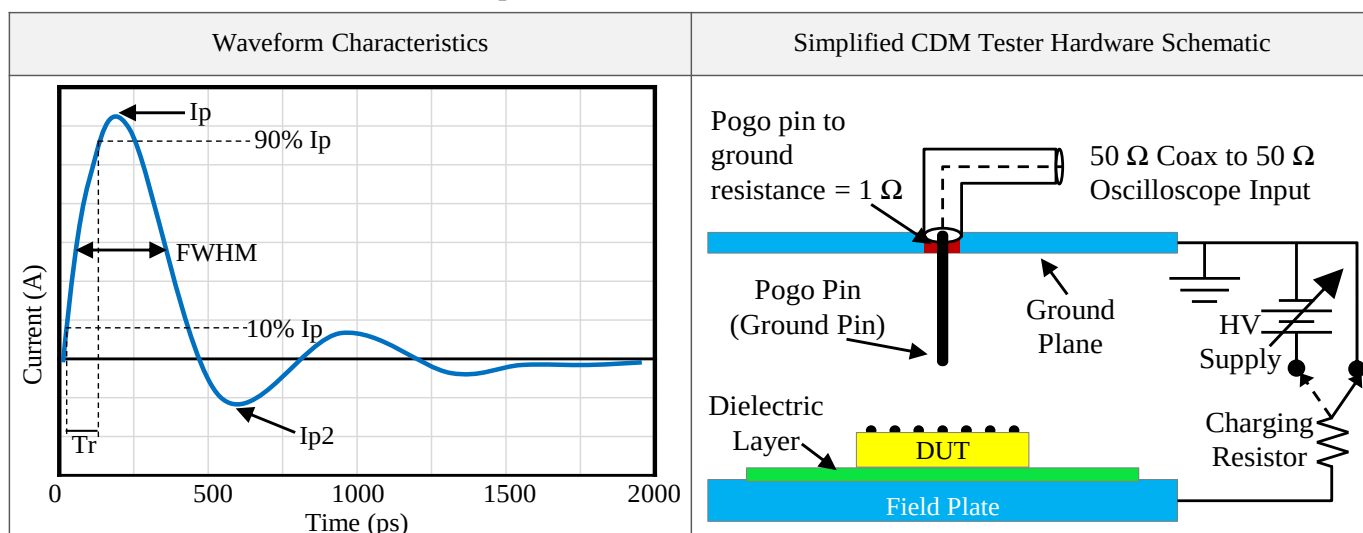
9.2 CDM (Charged Device Model)

This test is classifying microcircuits according to their susceptibility to damage or degradation by exposure to a defined electrostatic Charged Device Model (CDM) discharge (ESD).

9.2.1 CDM ESDS Device Classification Levels

Classification Level	Classification Test Condition (in Volts)
C0a	< 125
C0b	125 to < 250
C1	250 to < 500
C2a	500 to < 750
C2b	750 to < 1000
C3	≥ 1000

9.2.2 CDM Current Waveform & Simplified CDM Tester Hardware Schematic



9.2.3 CDM Tester

Tester Model	Maker	Serial Number	Effective term of Calibration	Accredited Calibration Lab.
Orion3	Thermo Fisher Scientific	1706181	2025.04.24 ~ 2026.04.24	SICT(KC01-018)
Orion3	Thermo Fisher Scientific	2006173	2025.04.24 ~ 2026.04.24	

10. Descriptions of Latch-up Test

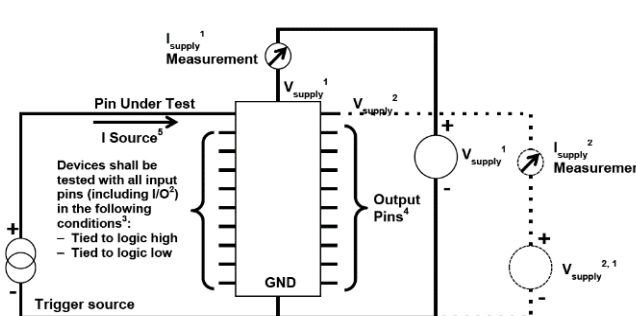
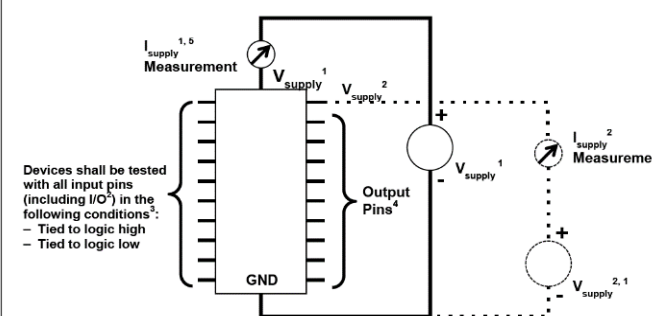
Latch-up pertains to a failure mechanism wherein a parasitic thyristor (such as a parasitic silicon controlled rectifier, or SCR) is inadvertently created within a circuit, causing a high amount of current to continuously flow through it once it is accidentally triggered or turned on. Depending on the circuits involved, the amount of current flow produced by this mechanism can be large enough to result in permanent destruction of the device due to electrical overstress (EOS).

10.1 Latch-up immunity levels

Immunity Level	Test Type	Magnitude of Trigger Force Current or Voltage
A	Positive Signal Pin Test ¹⁾	$\geq 100 \text{ mA}$
	Negative Signal Pin Test ¹⁾	The actual injected current may be less than 100 mA, if pin voltage preset limits are reached
	Supply Test	1.5 x VDD(Max.) or MSV, whichever is less The actual applied voltage may be less than 1.5 x V _{maxSUP} or MSV, if supply current limits are reached
B	If immunity level A cannot be achieved	

(1) In the Signal Pin Test, current injection is achieved either by current forcing with voltage compliance limit (I-Test) or by applying voltage with current compliance limit (E-Test).

10.2 The equivalent circuit for Latch-up Test

Signal Pin Test(I-Test or E-Test)	Supply Test
	

10.3 Latch-up Tester

Tester Model	Maker	Serial Number	Effective term of Calibration	Accredited Calibration Lab.
MK2	Thermo Fisher Scientific	401208	2025.10.22 ~ 2026.10.22	SICT(KC01-018)
MK4. TE	Thermo Fisher Scientific	1806237	2025.10.22 ~ 2026.10.22	

11. Test Pin Group

11.1 ESD (HBM, CDM) Test Pin Group

Pin Group	Type	Pin(s)
VPP	POWER	8
GND	GROUND	1
I/O	I/O	3,5,6
N/C	N/C	2,4,7

11.2 Latch-up Test Pin Group

Pin Group	Type	Pin(s)
VPP_5.5	POWER	8
GND	GROUND	1
I/O_5.5	I/O	3,5,6
N/C	N/C	2,4,7

MECHANICAL TEST REPORT [PRODUCT : UONE-A10]

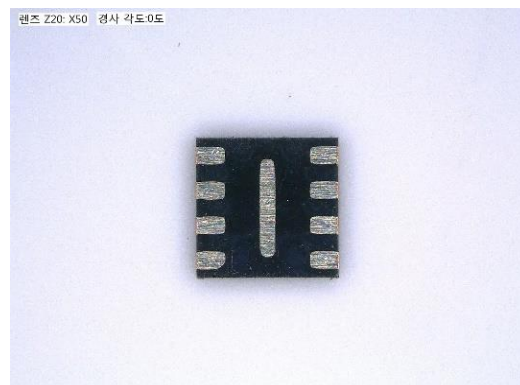
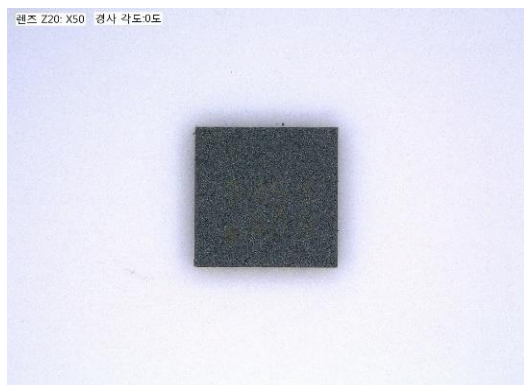
COMPANY	UBIVELOX Inc.	PRODUCT	UONE-A10	ISSUED DATE	25.11.26
Laboratory	QRT	THE START DATE	25.11.18	THE END DATE	25.11.19

TEST ITEM	TEST CONDITION	SAMPLE SIZE	TEST RESULT ¹⁾	REF. DOCUMENT
Solderability [Dip & Look]	Preconditioning : Condition E Dry Bake : 155 °C Exposure time : 4 hours ± 15 min Flux Immersion Time : 5 - 10 sec Solder Temperature : 245 ± 5 °C Solder Immersion Time : 5 +0/-0.5 sec Solder Immersion speed : 25 ± 6 mm/sec Solder Immersion depth : 0.5 ± 0.1 mm Solder Dipping angle : Horizontal	15 ea	PASS	J-STD-002D:2011

Note..

- 1) Accept/Reject Criteria All leads shall exhibit a continuous solder coating free from defects for a minimum of 95% of the critical area of any individual lead.

PRODUCT IMAGE



MECHANICAL TEST REPORT [PRODUCT : UONE-A10]

Solderability [Dip & Look]

TEST EQUIPMENT



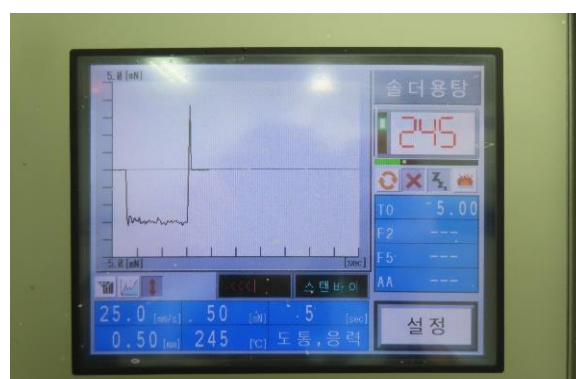
Equipment Name : HTS-16
Equipment Maker : JEIOTECH
Equipment Model : HB-502SS
Validity Period : 2025. 10. 16 ~ 2026. 10. 16



Flux Name : IPC J-STD-002, Flux #2
Flux type : RMA Liquid Flux
Colophony : 25% +/- 0.5%
Diethylammonium hydrochloride (CAS 660-68-4) : 0.39% +/- 0.01%
Isopropyl alcohol : 74.61% +/- 0.5%



Equipment Name : Solderability
Equipment Maker : RHESCA
Equipment Model : SAT-5200
Solder Type : Sn 96.5/ Ag 3/ Cu 0.5 (Pb-free RoHS Solder)

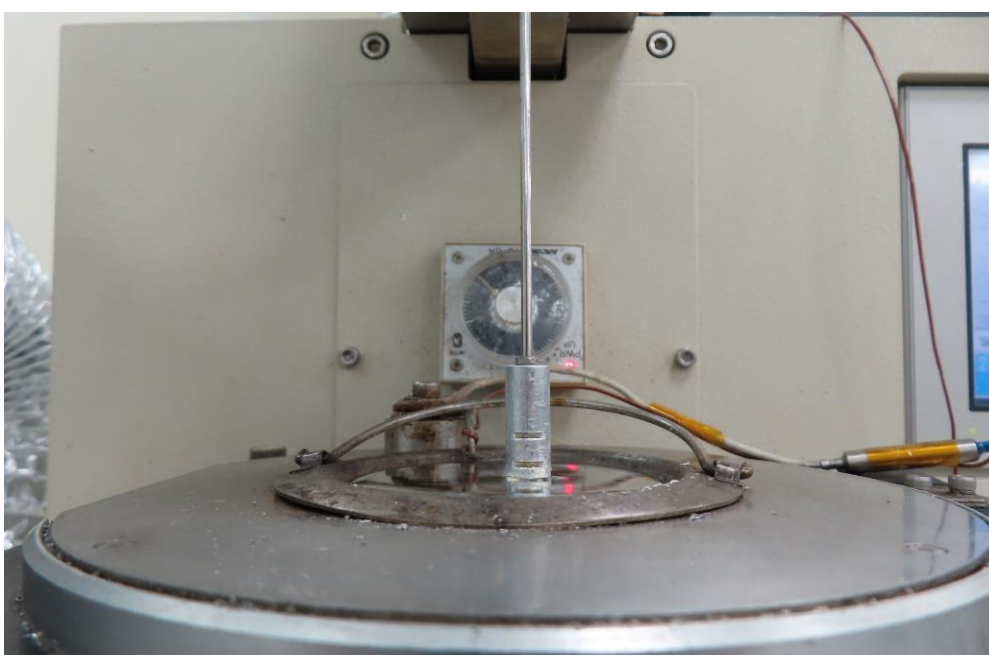


Test Profile

MECHANICAL TEST REPORT [PRODUCT : UONE-A10]

Solderability [Dip & Look]

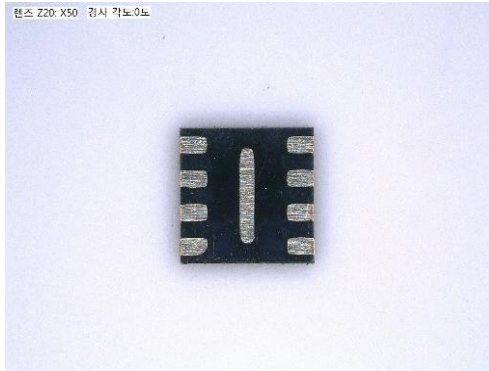
TEST IMAGE



MECHANICAL TEST REPORT [PRODUCT : UONE-A10]

Solderability [Dip & Look]

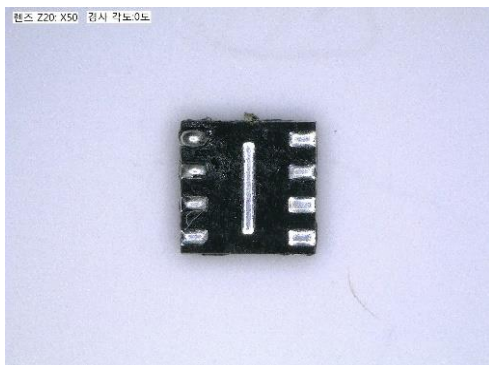
AFTER TEST IMAGE



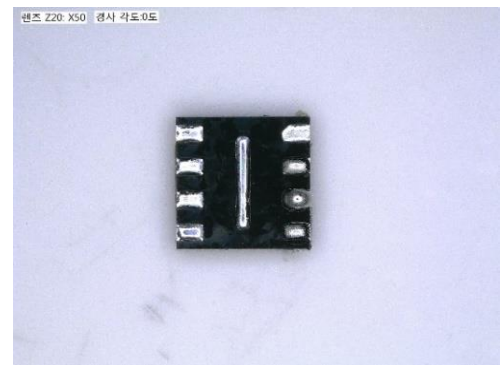
Before Test



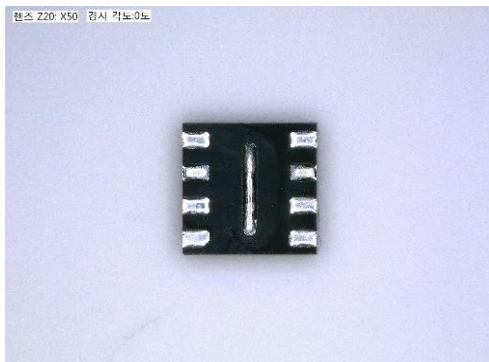
After Test 1



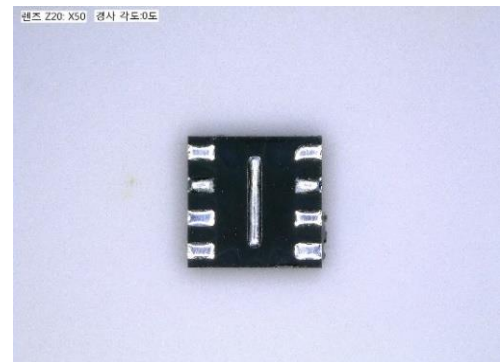
After Test 2



After Test 3



After Test 4

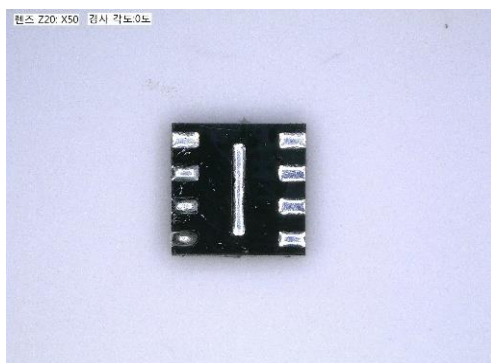


After Test 5

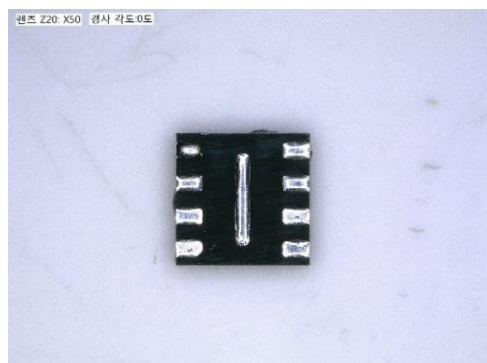
MECHANICAL TEST REPORT [PRODUCT : UONE-A10]

Solderability [Dip & Look]

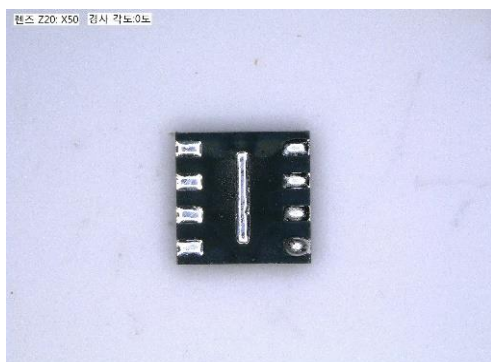
AFTER TEST IMAGE



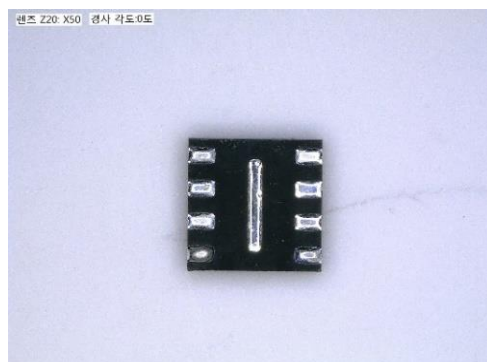
After Test 6



After Test 7



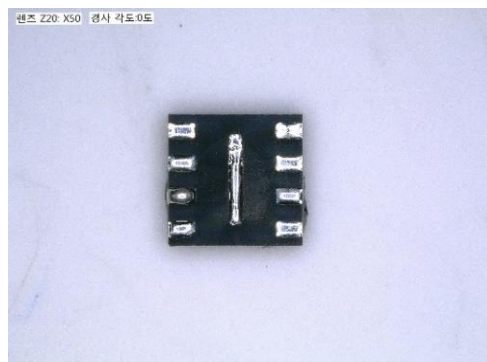
After Test 8



After Test 9



After Test 10

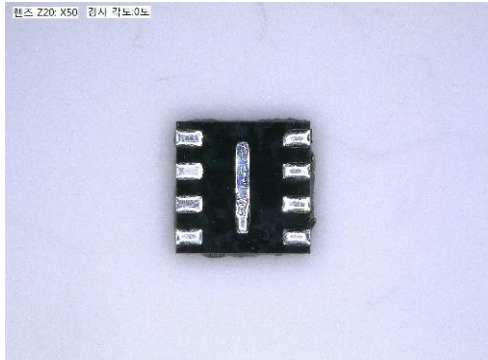


After Test 11

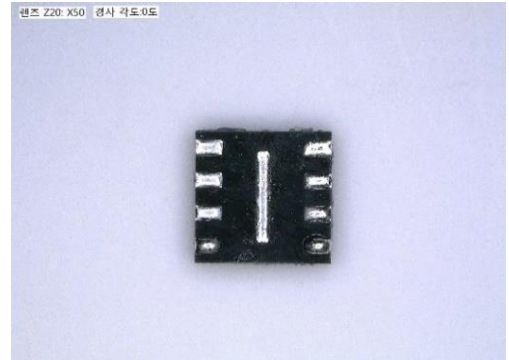
MECHANICAL TEST REPORT [PRODUCT : UONE-A10]

Solderability [Dip & Look]

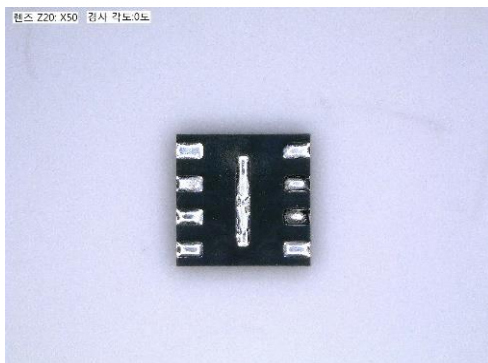
AFTER TEST IMAGE



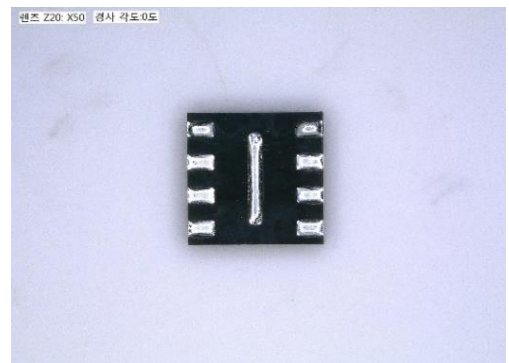
After Test 12



After Test 13



After Test 14



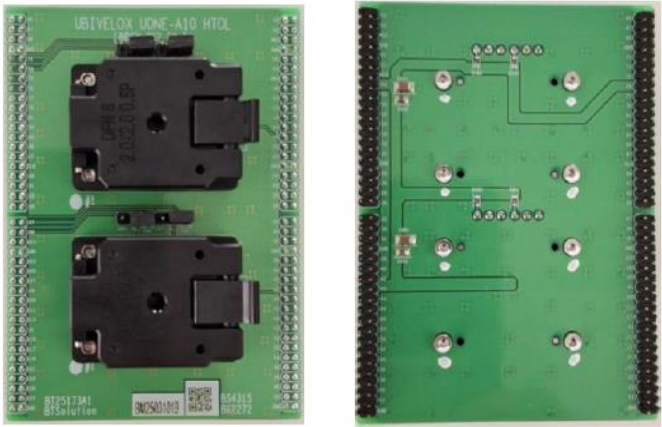
After Test 15

1. APPENDIX A: HTOL Setup Details

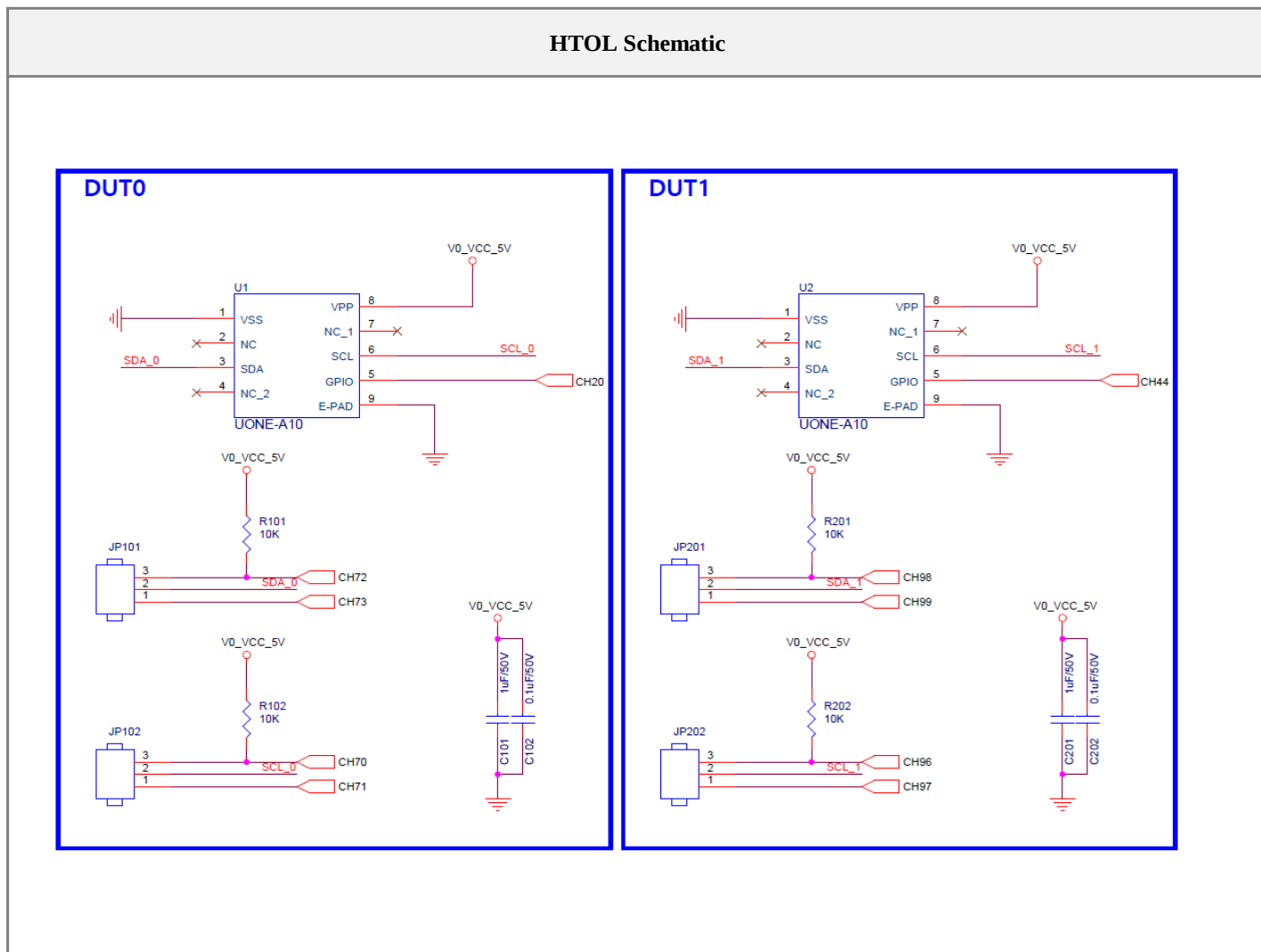
1.1 HTOL Test Configuration

Test Item : HTOL	Test condition : °C
Power Name	V0 : 5.5 V
VIH	3.6 V
PNP Name : UONE-A10	Cycle Time : 1000ns
Power up sequence	V0_5.5V
Power off sequence	V0_5.5V
Note	
- GPIO(RZ) Start – 500ns / Stop – 1000ns - SCL,SDA = Floating 처리	

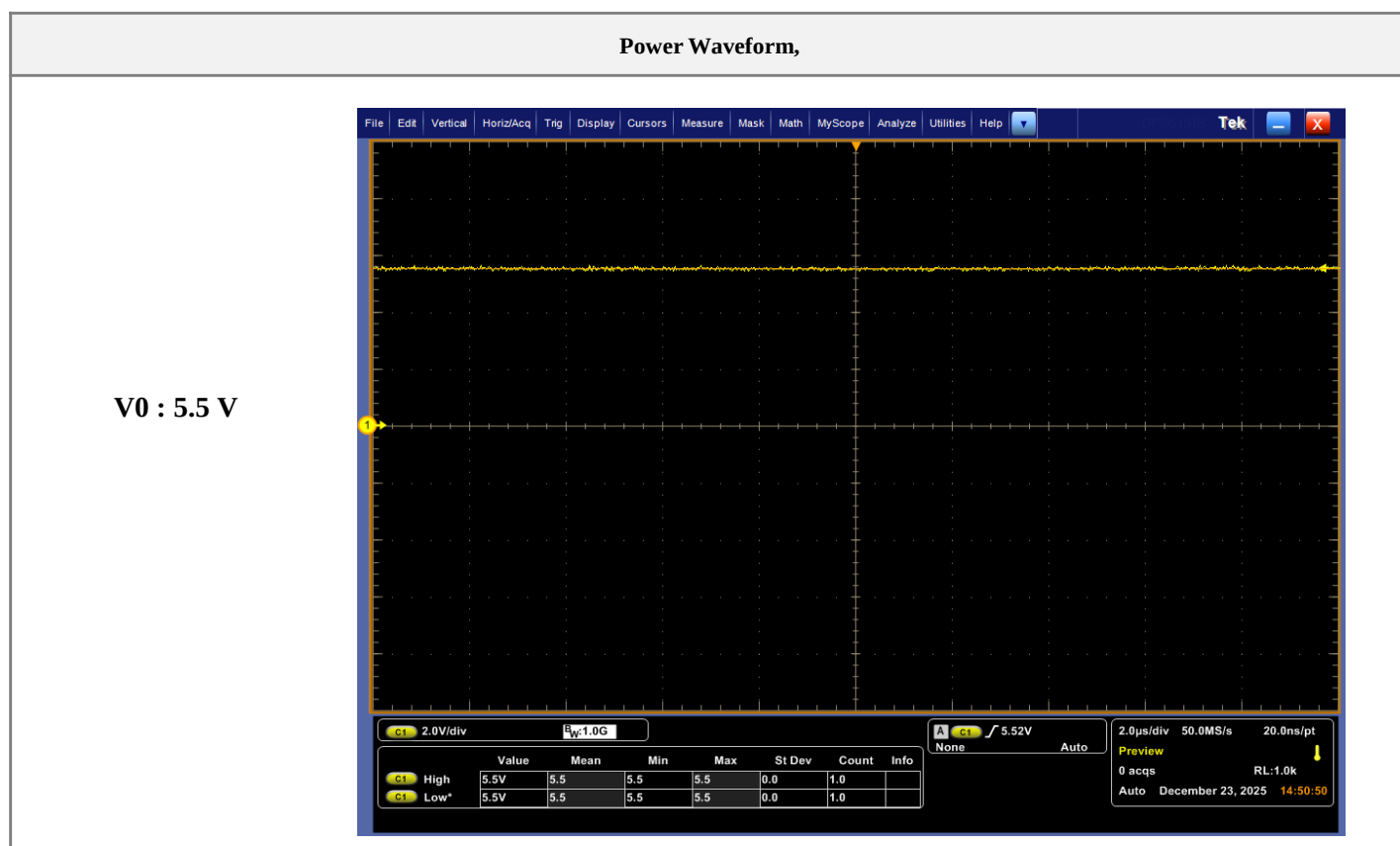
1.1.1 HTOL Test Board

Test Board	Board Information
	Product : UONE-A10
	Package Type : 80DFN
	Board Type : HTOL Socket Board
	Board S/N : BT25173A1
	Material Code : BS4315
	Quantity : 40 EA

1.1.2 HTOL Schematic



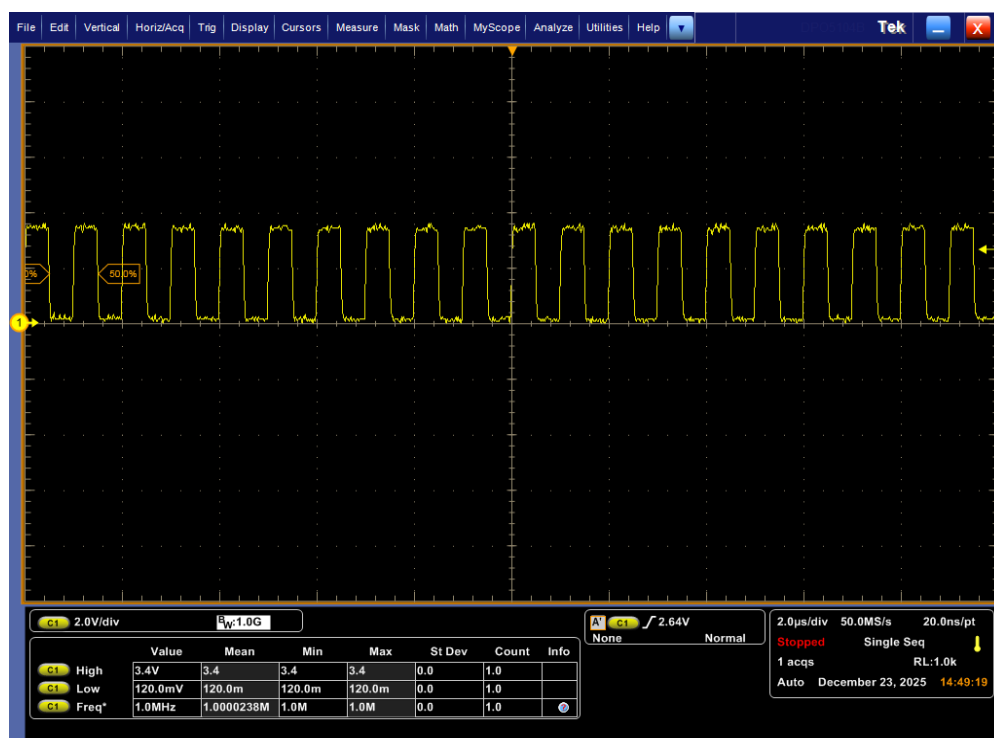
1.1.3 Operation Check



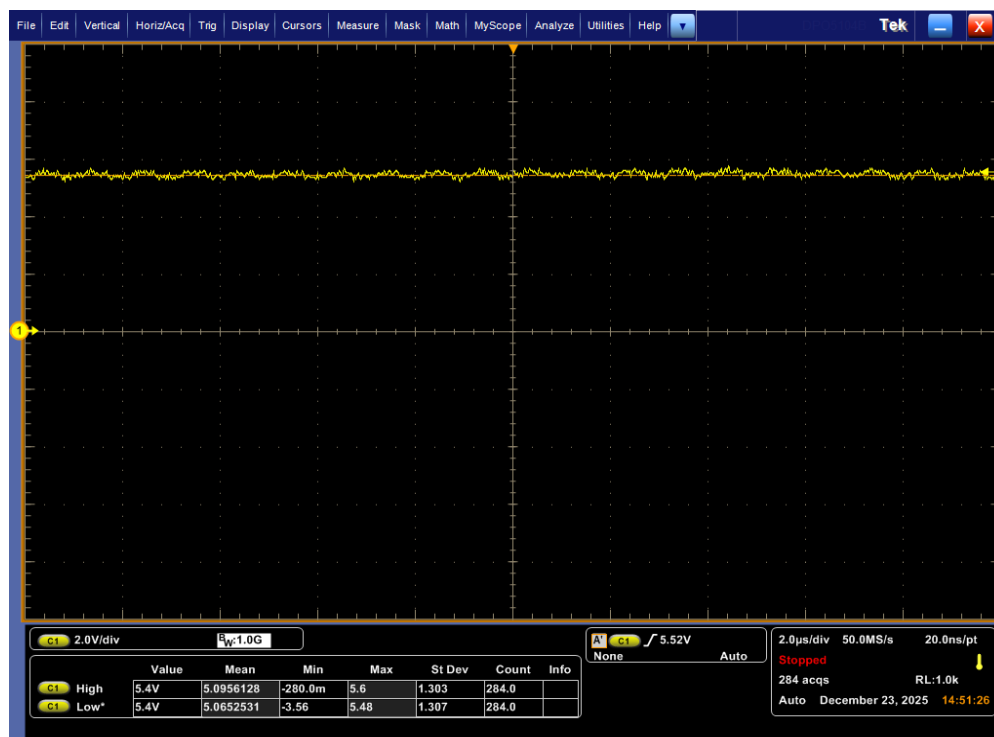
1.1.3 Operation Check

Input Waveform,

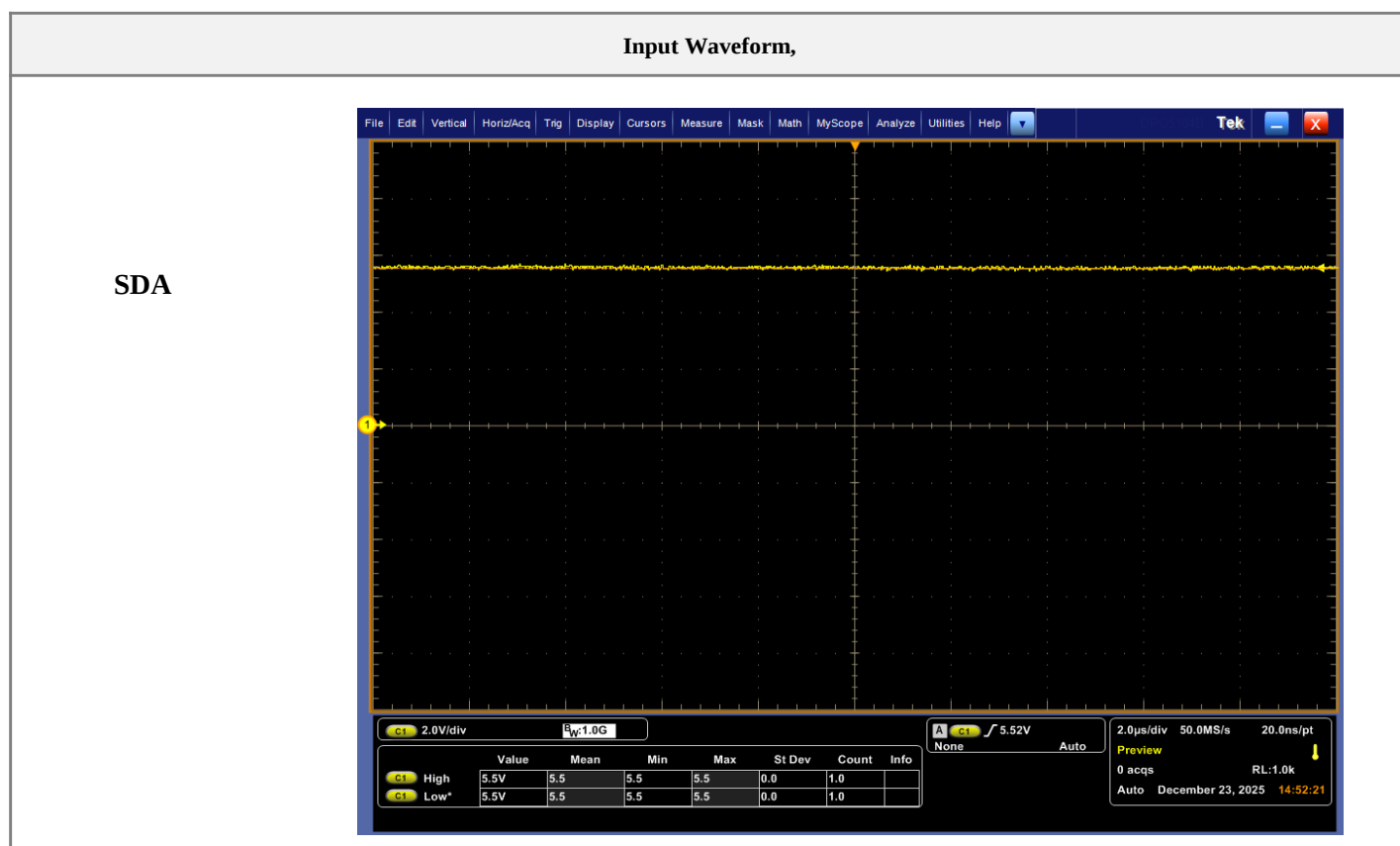
GPIO



SCL



1.1.3 Operation Check





1.1.3 Operation Check

Ambient	Qty	Power Name	Bias Set	Measured Current	Remark
Room	1	V0	5.5 V	0.001 A	
Room condition_1ea					
SLOT 3	V0	V1	V2	V3	V4
	5.500	0.000	0.000	0.000	0.000
	5.502	-0.001	-0.001	-0.003	-0.002
	0.001	0.004	0.003	0.007	0.001
					Set Voltage
					Read Voltage
					Read Current



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